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Thermodynamic assessment of the Eu–Pb and Lu–Pb systems

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ABSTRACT

The thermodynamic assessment of the Eu–Pb and Lu–Pb binary systems has been carried out by using the Calculation of Phase Diagrams (CALPHAD) method based on the available literature data including the phase equilibria and thermodynamic properties. The Gibbs free energies of the liquid, bcc, fcc and hcp phases were described by the subregular solution model with Redlich–Kister formula, and those of the intermetallic compounds (Eu_2Pb, Eu_5Pb_3, βEuPb, αEuPb, $EuPb_3$, Lu_5Pb_3, βLu_5Pb_4, αLu_5Pb_4, Lu_6Pb_5, and $LuPb_2$) in the binary systems were described by the two-sublattice model. A consistent set of the thermodynamic parameters leading to a reasonable agreement between the calculated results and literature data was obtained.

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1. Introduction

Mg base alloys with good performances are not only used as structural materials and hydrogen storage materials but also used extensively as anode materials in seawater battery systems [1–4]. Major attractions of Mg anode materials (e.g. Mg–Pb) are rapid activation, high cell voltage and wide voltage range, high power density capability, relatively light mass in inactivated state, and long inactivated storage life [5,6]. In order to further develop high-performance Mg anode materials, addition of rare earth elements (RE) is of both technical interest and industrial demand. Thus, accurate information on the phase diagrams of Mg base alloys with RE, which can not only help to effectively select the alloying method and heat treatment process but also help to well understand the relationship between the composition and the microstructure, is especially required. The Calculation of Phase Diagrams (CALPHAD) technique, which has been recognized as an important tool to significantly reduce time and cost during the development of materials, can provide a clear guidance for the materials design [7–9]. Our goal is to develop a thermodynamic database of Mg–Pb–Hg–RE alloy systems for reliable predictions of liquids, phase fraction, equilibrium and non-equilibrium solidification behavior, etc. in multi-component systems, and has made some thermodynamic assessments in the RE alloy systems [10–15]. In present work, as a part of the above-mentioned thermodynamic database, the thermodynamic assessment of the Eu–Pb and Lu–Pb binary systems were carried out by means of the CALPHAD method.

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2. Literature data

2.1. Eu–Pb system

Regarding to the information on phase equilibria and phase transformations, Mcmasters and Gscheidner [16] presented the phase diagram of the Eu–Pb system (see Fig. 1), where two solution phases ((Eu), (Pb)) and five intermediate phases (Eu_2Pb, Eu_5Pb_3, βEuPb, αEuPb, and $EuPb_3$) were confirmed. In addition, three eutectic reactions ($L \rightarrow Eu_2Pb + (Eu)$), ($L \rightarrow Eu_5Pb_3 + \beta EuPb$) and ($L \rightarrow \alpha EuPb + EuPb_3$) and their eutectic points (770 °C and 5.5 at% Pb), (1031 °C and 44.5 at% Pb) and (717 °Cand 63.5 at% Pb), were also reported, respectively.

Regarding to the thermodynamic properties, the enthalpy of formation of the $EuPb_3$ phases at 25 °C was measured by Palenzona and Cirafici [17], and the enthalpy of formation of the $EuPb_3$ phase at 393–750 °C was reported by Kober et al. [18].

2.2. Lu–Pb system

Regarding to the information on phase equilibria and phase transformations, Mcmasters and Gschneidner [19] presented the phase diagram of the Pb–Lu system (see Fig. 2), where two solution phases ((Lu), (Pb)) and five intermediate phases (Lu_5Pb_3, βLu_5Pb_4, αLu_5Pb_4, Lu_6Pb_5, and $LuPb_2$) were confirmed, and Borzone et al. [21] mentioned that the compound $LuPb_3$ cannot be prepared at normal pressure and synthesized at high pressure (67–88 kbar) between 973 and 1373 K, which suggests that the $LuPb_3$ phase should be an unstable phase in the Lu–Pb phase diagram at normal pressure. Thus, the $LuPb_3$ phase was not taken into account in the present assessment. In addition, the eutectic

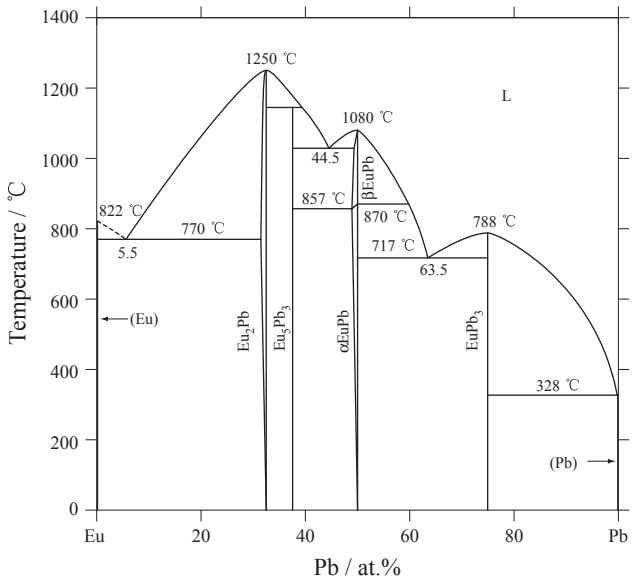

Fig. 1. The phase diagram of the Eu–Pb system reviewed by Mcmasters and Gscheidner [16].

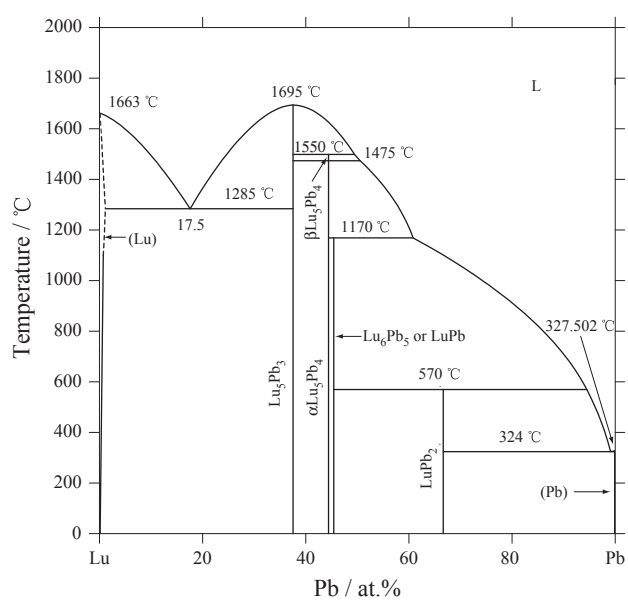

Fig. 2. The phase diagram of the Lu–Pb system reviewed by Mcmasters and Gschneidner [19].

Table 1
The stable solid phases and the used models in the Eu–Pb system.

Phase	Composition (at% Pb)	Prototype	Strukturbericht symbol	Modeling phase	Used models
(Eu)	0	W	Bcc	$(Eu, Pb)_1$	SSM
Eu_2Pb	33–33.3	Co_2Si	C23	$(Eu, Pb)_{0.667}(Pb)_{0.333}$	TSM
Eu_5Pb_3	37.5	W_5Si_3	$D8_m$	$(Eu)_{0.625}(Pb)_{0.375}$	SM
βEuPb	50	–	–	$(Eu)_{0.5}(Eu, Pb)_{0.5}$	TSM
αEuPb	50	AuCu	$L1_0$	$(Eu)_{0.5}(Eu, Pb)_{0.5}$	TSM
$EuPb_3$	75	$AuCu_3$	$L1_2$	$(Eu)_{0.25}(Pb)_{0.75}$	SM
(Pb)	100	Cu	Fcc	$(Eu, Pb)_1$	SSM

SSM: subregular solution model, SM: sublattice model and TSM: two-sublattice model.

Table 2
The stable solid phases and the used models in the Lu–Pb system.

Phase	Composition (at% Pb)	Prototype	Strukturbericht symbol	Modeling phase	Used models
(Lu)	0–1	Mg	Hcp	$(Lu, Pb)_1$	SSM
Lu_5Pb_3	37.5	Mn_5Si_3	$D8_8$	$(Lu)_{0.625}(Pb)_{0.375}$	SM
βLu_5Pb_4	44.4	–	–	$(Lu)_{0.556}(Pb)_{0.444}$	SM
αLu_5Pb_4	44.4	–	–	$(Lu)_{0.556}(Pb)_{0.444}$	SM
Lu_6Pb_5	45.5	–	–	$(Lu)_{0.545}(Pt)_{0.455}$	SM
$LuPb_2$	66.7	$MoSi_2$	$C11_b$	$(Lu)_{0.333}(Pb)_{0.667}$	SM
(Pb)	100	Cu	Fcc	$(Lu, Pb)_1$	SSM

SSM: subregular solution model and SM: sublattice model.

Table 3
The optimized thermodynamic parameters of the Pb–Eu system.

Parameters in each phase (J/mol)
Liquid phase, format $(Eu, Pb)_1$
${}^0L^{Liq}_{Eu,Pb} = -1{,}15{,}500 + 17.6T$
${}^1L^{Liq}_{Eu,Pb} = 32{,}000 - 3T$
${}^2L^{Liq}_{Eu,Pb} = 20{,}000 - 5T$
Bcc phase, format $(Eu, Pb)_1$
${}^0L^{Bcc}_{Eu,Pb} = 0$
Eu_2Pb phase, format $(Eu)_{0.667}(Eu, Pb)_{0.333}$
${}^0G^{Eu_2Pb}_{Eu:Pb} = -26{,}450 - 2.35T + 0.667G^{Bcc}_{Eu} + 0.333G^{Fcc}_{Pb}$
${}^0G^{Eu_2Pb}_{Eu:Eu} = 5000 + G^{Bcc}_{Eu}$
${}^0G^{Eu_2Pb}_{Eu:Eu,Pb} = 4200$
Eu_5Pb_3 phase, format $(Eu)_{0.625}(Pb)_{0.375}$
${}^0G^{Eu_5Pb_3}_{Eu:Pb} = -28{,}050 - 2.350T + 0.625{}^0G^{Bcc}_{Eu} + 0.375{}^0G^{Fcc}_{Pb}$
βEuPb phase, format $(Eu)_{0.5}(Eu, Pb)_{0.5}$
${}^0G^{\beta EuPb}_{Eu:Pb} = -30{,}400 - 3.14T + 0.5G^{Bcc}_{Eu} + 0.5G^{Fcc}_{Pb}$
${}^0G^{\beta EuPb}_{Eu:Eu} = 4000 + G^{Bcc}_{Eu}$
${}^0G^{\beta EuPb}_{Eu:Eu,Pb} = -15{,}000 + 8T$
αEuPb phase, format $(Eu)_{0.5}(Eu, Pb)_{0.5}$
${}^0G^{\alpha EuPb}_{Eu:Pb}{}^0 = -33{,}200 - 0.7T + 0.5G^{Bcc}_{Eu} + 0.5G^{Fcc}_{Pb}$
${}^0G^{\alpha EuPb}_{Eu:Eu} = 5000 + G^{Bcc}_{Eu}$
${}^0G^{\alpha EuPb}_{Eu:Eu,Pb} = -5000$
$EuPb_3$ phase, format $(Eu)_{0.25}(Pb)_{0.75}$
${}^0G^{EuPb_3}_{Eu:Pb} = -46{,}000 + 17T + 0.25{}^0G^{Bcc}_{Eu} + 0.75{}^0G^{Fcc}_{Pb}$
Fcc phase, format $(Eu, Pb)_1$
${}^0L^{Fcc}_{Eu} = 5000 + G^{Bcc}_{Eu}$
${}^0L^{Fcc}_{Eu,Pb} = 0$

reaction ($L \rightarrow Lu_5Pb_3 + (Pb)$) occurring at 1285 ± 5 °C and 17.5 ± 0.5 at% Pb was also reported.

Regarding to the thermodynamic properties, Witusiewicz et al. [20] provided the information on the enthalpy and entropy of formation of the $LuPb_2$ phase at 25–606 °C and the enthalpy of formation of the $LuPb_3$ phase at 25–727 °C, and Borzone et al. [21] measured the molar heat capacity of formation of the Lu_5Pb_3 phase, Lu_6Pb_5 phase and alloy with 97 at% Pb phases at 250–550 °C, and the electromotive forces of the galvanic cell for Lu–Pb alloys (Lu = 0.010, 0.015, 0.020, 0.030, 0.040) at 337–457 °C.

All the above-mentioned literature data including phase equilibria and thermodynamic properties will be used in the following thermodynamic assessment of the Eu–Pb and Lu–Pb systems.

Table 4
The optimized thermodynamic parameters of the Pb-Lu system.

Parameters in each phase (J/mol)
Liquid phase, format (Lu, Pb)$_1$
${}^0L^{Liq}_{Lu,Pb} = -1{,}10{,}500 + 8.0T$
${}^1L^{Liq}_{Lu,Pb} = -95{,}800 + 55T$
${}^2L^{Liq}_{Lu,Pb} = 83{,}000 - 40T$
${}^3L^{Liq}_{Lu,Pb} = 1{,}20{,}000 - 65T$
Hcp phase, format (Lu, Pb)$_1$
${}^oL^{Hcp}_{Lu,Pb} = -95{,}000 + 35T$
Lu_5Pb_3 phase, format $(Lu)_{0.625}(Pb)_{0.375}$
${}^0G^{Lu_5Pb_3}_{Lu:Pb} = -41{,}000 + 2.25T + 0.625\,{}^0G^{Hcp}_{Lu} + 0.375\,{}^0G^{Fcc}_{Pb}$
βLu_5Pb_4 phase, format $(Lu)_{0.556}(Pb)_{0.444}$
${}^0G^{\beta Lu_5Pb_4}_{Lu:Pb} = -40{,}640 + 1.66T + 0.556G^{Hcp}_{Lu} + 0.444G^{Fcc}_{Pb}$
αLu_5Pb_4 phase, format $(Lu)_{0.556}(Pb)_{0.444}$
${}^0G^{\alpha Lu_5Pb_4}_{Lu:Pb} = -44{,}538 + 3.89T + 0.556G^{Hcp}_{Lu} + 0.444G^{Fcc}_{Pb}$
Lu_6Pb_5 phase, format $(Lu)_{0.545}(Pb)_{0.455}$
${}^0G^{Lu_6Pb_5}_{Lu:Pb} = -44{,}540 + 4.150T + 0.545\,{}^0G^{Hcp}_{Lu} + 0.455\,{}^0G^{Fcc}_{Pb}$
$LuPb_2$ phase, format $(Lu)_{0.333}(Pb)_{0.667}$
${}^0G^{LuPb_2}_{Lu:Pb} = -34{,}200 + 9.65T + 0.333\,{}^0G^{Hcp}_{Lu} + 0.667\,{}^0G^{Fcc}_{Pb}$
Fcc phase, format (Lu, Pb)$_1$
${}^0L^{Fcc}_{Lu} = 5000 + G^{Hcp}_{Lu}$
${}^0L^{Fcc}_{Lu,Pb} = 0$

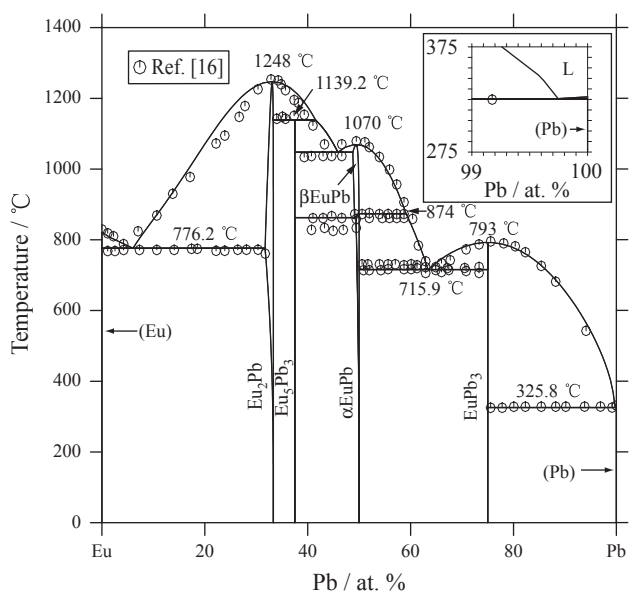

Fig. 3. The calculated phase diagram of the Eu–Pb system compared with the experimental data [16].

3. Thermodynamic models

The related information of stable solid phases and the models used in the Eu–Pb and Lu–Pb systems are listed in Tables 1 and 2, respectively.

3.1. Solution phases

In the Eu–Pb and Lu–Pb binary systems (RE–Pb system), Gibbs free energies of the liquid, bcc, fcc, and hcp are described by the subregular solution model with Redlich–Kister formula [22]. The

Table 5
The calculated invariant reactions in the Pb–Eu system compared with the literature data [16].

Reaction	Temperature (°C)	Composition (at% Pb)			References
L↔(Eu)+Eu_2Pb	770 ± 2	5.5 ± 0.2	0	33	[16]
	776	6.02	0	31.7	This work
L↔Eu_2Pb	1250 ± 3	33.3		33.3	[16]
	1247.6	33.3		33.3	This work
Eu_2Pb+L↔Eu_5Pb_3	1145 ± 3	33.3		37.5	[16]
	1139.2	33.3	41.5	37.5	This work
L↔Eu_5Pb_3+βEuPb	1031 ± 2	44.5 ± 0.2	37.5	50	[16]
	1047.7	45.9	37.5	50	This work
βEuPb↔Eu_5Pb_3+αEuPb	857	50	37.5	550	[16]
	862.7	49.1	37.5	49.4	This work
L↔βEuPb	1080 ± 2	50		50	[16]
	1070	50		50	This work
βEuPb+L↔αEuPb	870 ± 3	50		50	[16]
	874	50	59.1	50	This work
L↔αEuPb+$EuPb_3$	717 ± 2	63.5 ± 0.2	50	75	[16]
	715.9	63.4	50	75	This work
L↔$EuPb_3$	788	75		75	[16]
	792.5	75		75	This work

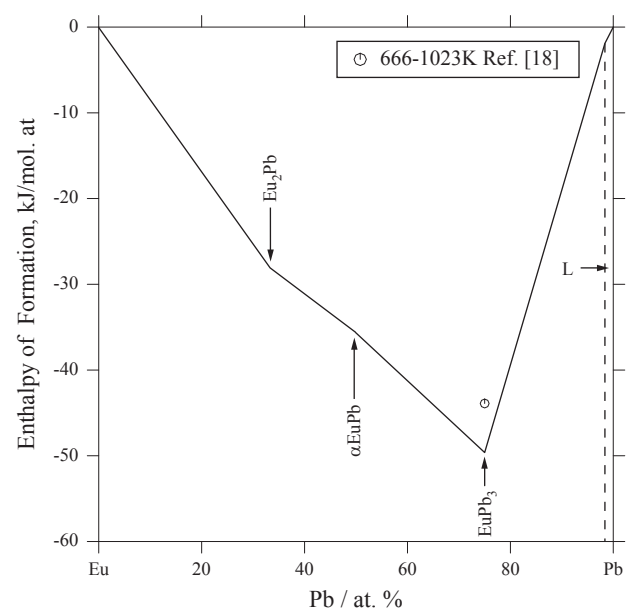

Fig. 4. The calculated standard enthalpies of formation with reference states of Eu (bcc) and Pb (fcc) in Eu–Pb system at 427 °C compared with the experimental data [18].

molar Gibbs free energies of solution phase in the RE–Pb system is given as follows:

$$G^{\phi}_m = \sum_{i=RE,Pb} {}^0G^{\phi}_i x_i + RT \sum_{i=RE,Pb} x_i \ln x_i + x_{RE}x_{Pb} \sum_{j=0}^{n} {}^jL^{\phi}_{RE,Pb}(x_{RE} - x_{Pb})^j \quad (1)$$

where ${}^0G^{\phi}_i$ is the Gibbs free energy of the pure component i in the respective reference state with the ϕ phase, which is taken from the SGTE pure element database [23], x_i represents the mole fraction of the component i; R is the gas constant and T is the absolute temperature; ${}^jL^{\phi}_{RE,Pb}$ is the binary interaction parameter and is expressed as

$${}^jL^{\phi}_{RE,Pb} = a_j + b_jT \quad (2)$$

where the values of a_j and b_j were evaluated in the present work based on the literature data.

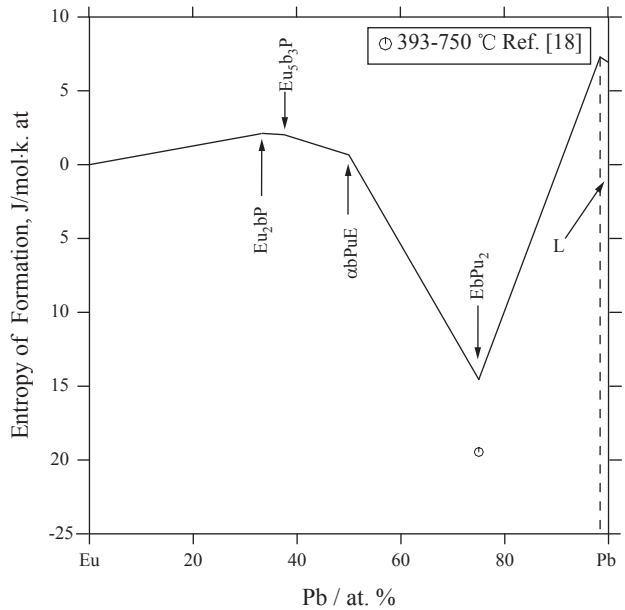

Fig. 5. The calculated standard entropy of formation with reference states of Eu (bcc) and Pb (fcc) in Eu–Pb system at 427 °C compared with the experimental data [18].

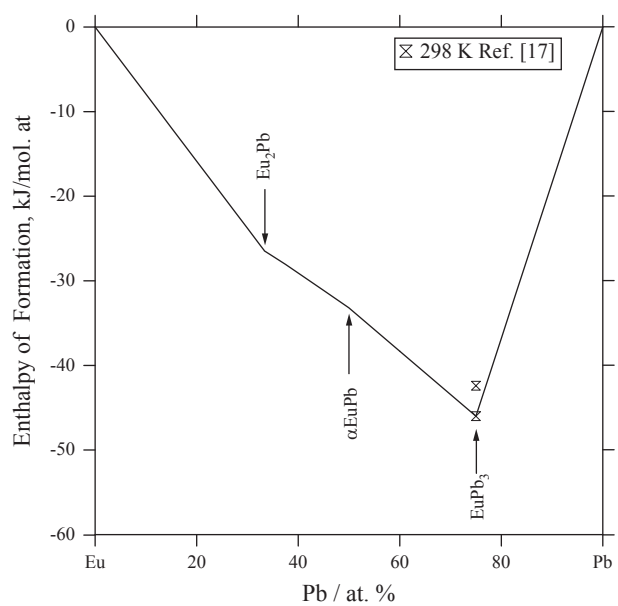

Fig. 6. The calculated standard enthalpies of formation with reference states of Eu (bcc) and Pb (fcc) in Eu–Pb system at 27 °C compared with the experimental data [17].

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds (Eu_5Pb_3, $EuPb_3$, Lu_5Pb_3, βLu_5Pb_4, αLu_5Pb_4, Lu_6Pb_5, $LuPb_2$) in the RE–Pb binary system are treated as the stoichiometric compound. The Gibbs free energy per molar of formula unit $(RE)_m(Pb)_n$ can be expressed as follows:

$$G_{RE:Pb}^{RE_mPb_n} = \Delta^0 G_f^{RE_mPb_n} + m^0 G_{RE}^{ref} + n^0 G_{Pb}^{ref} \quad (3)$$

where the $\Delta^0 G_f^{RE_mPb_n}$ indicate the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is formulated as

$$\Delta^0 G_f^{RE_mPb_n} = a_1 + b_1 T \quad (4)$$

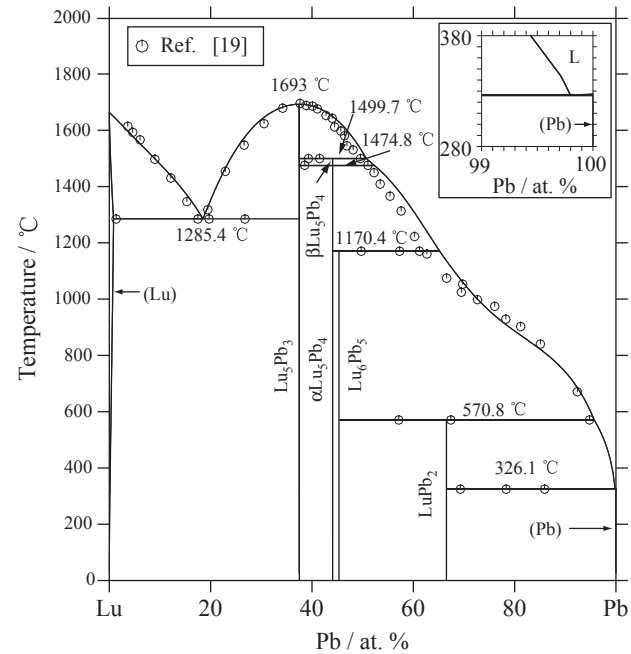

Fig. 7. The calculated phase diagram of the Lu–Pb system compared with the experimental data [19].

Table 6
The calculated invariant reactions in the Lu–Pb system compared with the literature data [19].

Reaction	Temperature (°C)	Composition (at% Pb)			References
$L \leftrightarrow (Lu) + Lu_5Pb_3$	1285 ± 5	17.5 ± 0.5	1	37.5	[19]
	1281.7	18.6	0.8	37.5	This work
$L \leftrightarrow Lu_5Pb_3$	1695 ± 10	37.5		37.5	[19]
	1700	37.5		37.5	This work
$Lu_5Pb_3 + L \leftrightarrow \beta Lu_5Pb_4$	1500 ± 10	37.5		44.4	[19]
	1501.25	37.5	50.6	44.4	This work
$\beta Lu_5Pb_4 + L \leftrightarrow \alpha Lu_5Pb_4$	1475 ± 10	44.4		44.4	[19]
	1475.7	44.4	52.2	44.4	This work
$L + \alpha Lu_5Pb_4 \leftrightarrow Lu_6Pb_5$	1170 ± 10				[19]
	1172.3	64.6	44.4	45.5	This work
$L + Lu_6Pb_5 \leftrightarrow LuPb_2$	570 ± 5				[19]
	570.2	94.8	45.5	66.7	This work
$L \leftrightarrow LuPb_2 + (Pb)$	324				[19]
	325	99.7	66.7	100	This work

where the parameters of a_1 and b_1 were evaluated in the present work.

3.3. Non-stoichiometric intermetallic phases

The Eu_2Pb, βEuPb, and αEuPb phases in the Eu–Pb binary system were treated as a non-stoichiometric intermetallic compound. The Gibbs free energy of the Eu_2Pb, βEuPb and αEuPb phases was described by the two-sublattice model [24]. The Gibbs free energy of formation per mole of formula unit $(Eu, Pb)_m(Eu, Pb)_n$ can be expressed as the following equation with reference to the pure elements in their non-magnetic state:

$$G_m^{(Eu,Pb)_m(Eu,Pb)_n} = y_{Eu}^I y_{Eu}^{II}\,{}^0G_{Eu:Eu} + y_{Eu}^I y_{Pb}^{II}\,{}^0G_{Eu:Pb} + y_{Pb}^I y_{Eu}^{II}\,{}^0G_{Pb:Eu} + y_{Pb}^I y_{Pb}^{II}\,{}^0G_{Pb:Pb} + mRT(y_{Eu}^I \ln y_{Eu}^I + y_{Pb}^I \ln y_{Pb}^I) + nRT(y_{Eu}^{II} \ln y_{Eu}^{II} + y_{Pt}^{II} \ln y_{Pt}^{II}) + m y_{Eu}^I y_{Pb}^I \left[y_{Eu}^{II} \sum_n {}^nL_{Eu,Pb:Eu}(y_{Eu}^I - y_{Pb}^I)^n \right.$$

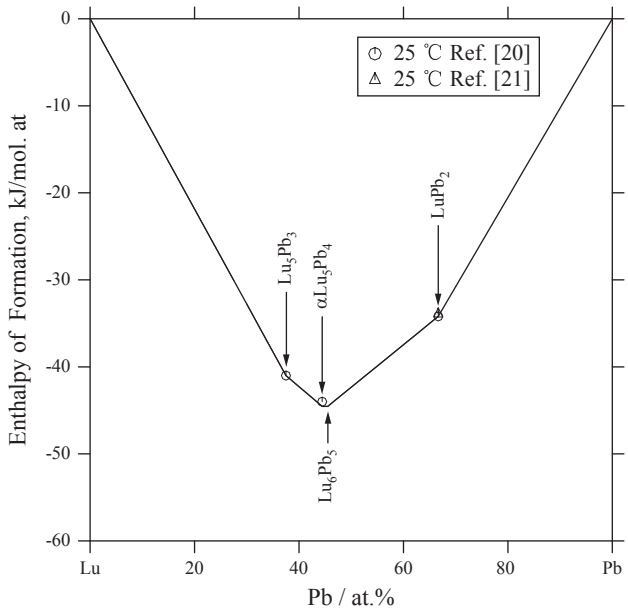

Fig. 8. The calculated standard enthalpies of formation with reference states of Lu (hcp) and Pb (fcc) in the Lu–Pb system at 25 °C compared with the literature data [20,21].

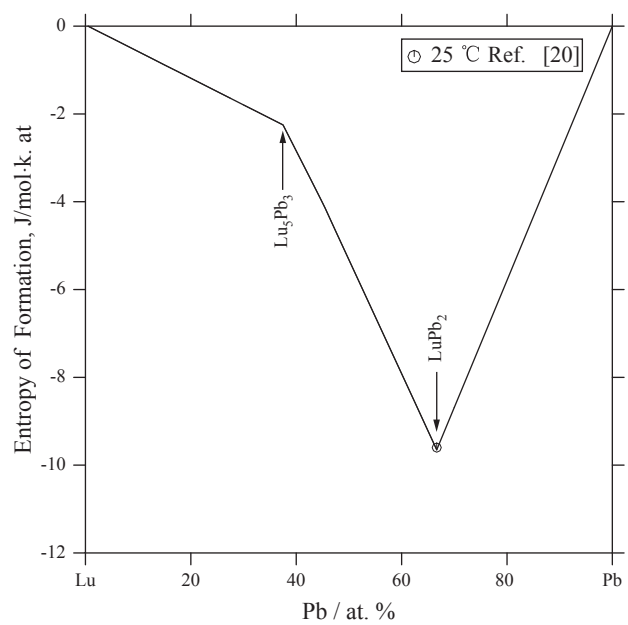

Fig. 9. The calculated standard entropy of formation with reference states of Lu (hcp) and Pb (fcc) in the Lu–Pb system at 25 °C compared with the literature data [20].

$$+y_{Pb}^{II}\sum_n {}^nL_{Eu,Pb:Pb}(y_{Eu}^{I}-y_{Pb}^{I})^n\Bigg]+ny_{Eu}^{II}y_{Pb}^{II}\Bigg[y_{Eu}^{I}\sum_n {}^nL_{Eu:Eu,Pb}(y_{Eu}^{II}-y_{Pb}^{II})^n$$

$$+y_{Pb}^{I}\sum_n {}^nL_{Pb:Eu,Pb}(y_{Eu}^{II}-y_{Pb}^{II})^n\Bigg] \quad (5)$$

where y_i^I and y_j^{II} are the site fractions of components i and j (i, j=Eu, Pb) located on sublattice I and II, respectively, and the parameter ${}^0G_{i:j}$ represents the Gibbs free energy of the compound phase when the two sublattices are occupied by element i or j. $L_{i:Eu,Pb}$ and $L_{Eu,Pb:j}$ are the interaction parameters between Eu and Pb in the second or first sublattice, when the other sublattice is occupied by element i or j. ${}^0G_{i:j}$, $L_{i:Eu,Pb}$ and $L_{Eu,Pb:j}$ were evaluated in the present work.

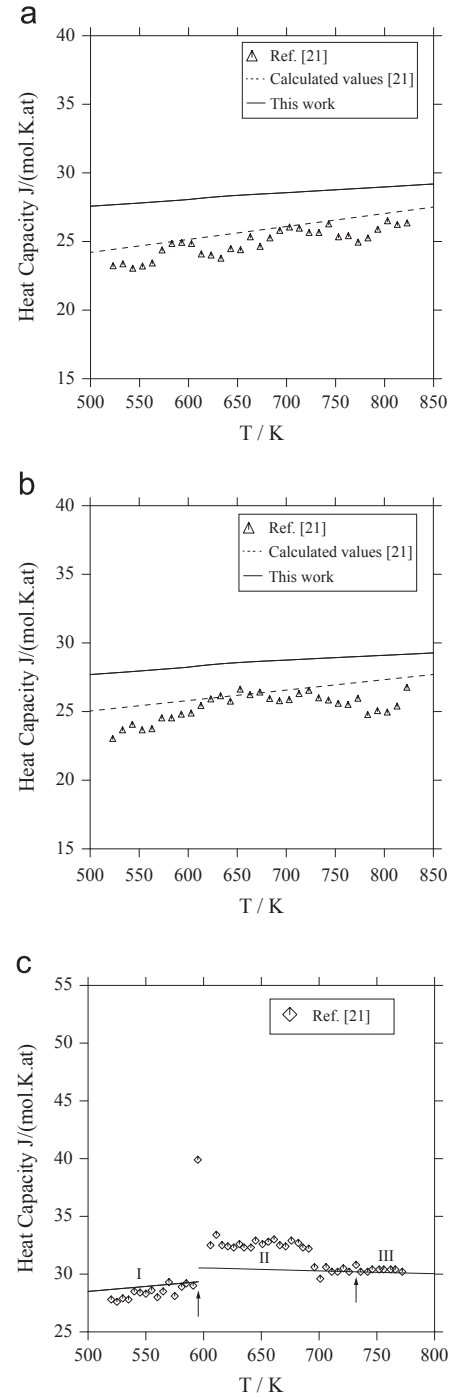

Fig. 10. The calculated molar heat capacity of formation of (a) the Lu_5Pb_3 phase, (b) the Lu_6Pb_5 phase, (c) alloy with 97 at% Pb in the Lu–Pb system compared with the literature data [21]. Especially, the arrows indicate the eutectic and liquidus temperatures. (Reference states, line I: Lu (hcp) and Pb (fcc), line II: Lu (hcp) and Pb (liquid), line III: Lu (liquid) and Pb (liquid)).

4. Results and discussion

The thermodynamic parameters of each phase in the Eu–Pb and Lu–Pb systems were optimized by both "manual" trial and error approach and PARROT [25] procedure. For the PARROT module in the THERMO-CALC [26] software, the literature data were used as input to the program. Each piece of selected

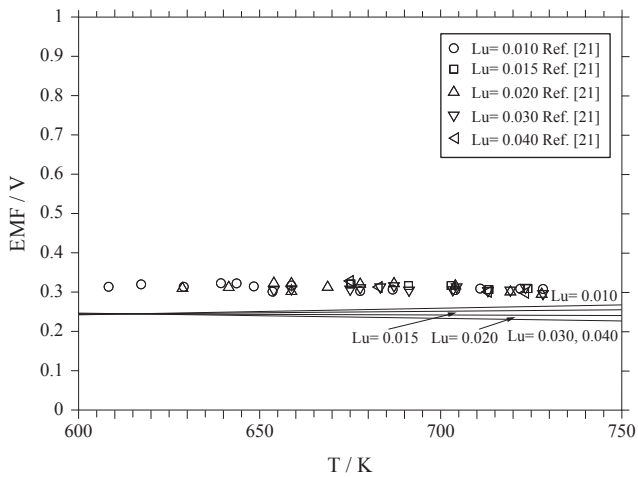

Fig. 11. The calculated electromotive forces of the galvanic cell with reference states of Lu (Hcp) and Pb (Liq) in Lu–Pb alloys (Lu=0.010, 0.015, 0.020, 0.030, 0.040) with the literature data [21].

information was given a certain weight by the importance of data, reflected in many points, e.g. experimental error, equipment precision, knowledge update and decade development, etc. In a word, the much more precise of data, the much more important of it, and changed by trial and error during the assessment, until most of the selected data is reproduced within the expected uncertainty limits. In the present work, all the mentioned literature data were used in the optimization process. All the parameters were eventually optimized to obtain the best consistency between the calculated results and literature data. All the optimized thermodynamic parameters of each phase in the Eu–Pb and Lu–Pb systems are listed in Tables 3 and 4, respectively.

4.1. Eu–Pb system

The calculated phase diagram of the Eu–Pb binary system compared with the literature data [16] is shown in Fig. 3. The calculated phase diagram is in good agreement with the phase diagram presented by Mcmasters and Gscheidner [16]. The calculated invariant reactions in the Eu–Pb system compared with experimental data [16] are listed in Table 5, where a reasonable agreement between the calculated and experimental temperatures for those invariant reactions was obtained.

The enthalpies and entropies of formation of the intermetallic compounds in the Eu–Pb system 427 °C were calculated, and presented in Figs. 4 and 5, respectively. The calculated results are consistent with the literature data in Ref. [18]. In addition, the calculated entropy at 27 °C in the Eu–Pb system are in good agreement with the reported data [17], as shown in Fig. 6.

4.2. Lu–Pb system

The calculated phase diagram of the Lu–Pb binary system compared with the literature data [19] is shown in Fig. 7. It can be seen that the calculated results are in reasonable agreement with the experimental data [19]. The calculated invariant reactions in the Lu–Pb system compared with experimental data [19] are listed in Table 6, where a reasonable agreement between the calculated and experimental temperatures for those invariant reactions was obtained.

The enthalpies and entropies of formation of the intermetallic compounds in the Lu–Pb system at 25 °C were calculated, and shown in Figs. 8 and 9, respectively. The calculated results are consistent with the literature data [20]. The molar heat capacity of the intermetallic compounds (Lu_5Pb_3, Lu_6Pb_5) and alloy with 97 at% Pb in the Lu–Pb system was calculated, and shown in Fig. 10. The calculated results are consistent with the literature data [21]. The trend of the calculation results compared with the experimental data is normal, according to calculated Gibbs free energy, thus we consider that the present results are reasonable.

The EMF data of the Lu–Pb alloys (Lu=0.010, 0.015, 0.020, 0.030, 0.040) at different temperatures were calculated through the following formula:

$$E = -\frac{\mu}{ZF} \tag{6}$$

where μ is the chemical potential of Lu in liquid phase, F is the Faraday constant and Z is the transferred electron number, is the outermost electrons of Lu. The calculated electromotive forces of the Lu–Pb binary system compared with the literature data [21] is shown in Fig. 11. When the content of Lu is above 0.020 at%, increase of temperature leads to decrease of the EMF data. The content of Lu is below 0.015 at%, the EMF data increase with temperature. Considering that there may be errors in experimental data, the calculated results are in reasonable agreement with the experimental data.

5. Conclusions

Based on the available literature data including the phase equilibria and thermodynamic properties, the thermodynamic asessment of the Eu–Pb and Lu–Pb binary systems has been carried out by using the CALPHAD technique. A consistent set of optimized thermodynamic parameters has been derived for describing the Gibbs free energy of each phase in the Eu–Pb and Lu–Pb binary systems, and a good agreement between the calculated results and most of the experimental data has been obtained.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2013.01.007.

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